

ITE 14 MIDTERM REVIEWER

I. Identification

1. A searching algorithm that checks each element from the beginning of the list until it finds a match or the whole list has been searched?

Answer: Linear Search

2. An ADT where elements could be added at the top of the list?

Answer: Stack

3. A computer programming technique involving the use of a procedure, subroutine, function, or algorithm that calls itself one or more times until a specified condition is met at which time the rest of each repetition is processed from the last one called to the first.

Answer: Recursion

4. It is a measurement of storage that an algorithm uses to execute properly?

Answer: Space Complexity

5. It is an amount of time to execute an algorithm ?

Answer: Time Complexity

6. Which is faster among the list: $T(n)$, $T(n^2)$, $T(n^3)$, $T(n \log n)$?

Answer: $T(n)$ is the fastest, followed by $T(n \log n)$, then $T(n^2)$, and finally $T(n^3)$. This ordering is based on their respective growth rates.

7. A structure containing a member that is a pointer to the same structure type?

Answer: Self-referential Structure

8. The _____ of a node is the number of edges from the root to the node.

Answer: depth

9. A function that calls itself either directly or indirectly through another function.

Answer: Recursive function

II. Comprehension

1. What is the running time of the code given below?

```
x = z[2] + 4 * x;
```

Answer: constant, denoted as $O(1)$

2. What is the running time of the code given below?

```
for (i=0; i<n; i++) {  
    y[i]=y[i]+i;  
    z = y[i]*2;  
}
```

Answer: the running time of the code is $O(n)$

3. What is the running time of the code given below?

```
for (i = 0 ; i<n ; i++)  
    for (j=0; j<i; j++)
```

```

if (a[j]<a[j+1]) {
temp=a[j];
a[j]=a[j+1];
a[j+1]=temp;
}

```

Answer: The calculation is as follows: The total number of operations = $(n * (n - 1)) / 2$. Therefore, the time complexity is $O(n^2)$.

4. In an array implementation of a queue where front holds the index value of the first added element and rear holds the index value of the last added element , if the initial value of front is - 1 for an empty queue, what is the value of rear if the queue has 1 element?

Answer: rear = 1

5. Find the upper bound for $f(n) = 5n^3 + n^2 + 8$. show your solution.

Answer: $O(n^3)$

The calculation for the solution is as follows:

- . Choose $g(n) = 5n^3$.
- . We need to find constants $c = 6$ and $n_0 = 1$.
- . Verify that $f(n) \leq cg(n)$ for all $n \geq n_0$:
 - For $n \geq 1$:
 - $1 + \frac{1}{5} + \frac{8}{5} = 1 + \frac{9}{5} < 6$.

Therefore, the upper bound for $f(n)$ is $O(n^3)$.

6. What is the value of *&ptrX?

```

int a = 10, *ptrX, *ptrY;
ptrX = &a; ptrY = &ptrX;
//Address of a is 1004, ptrX is 1008, and ptrY is 1012.

```

Answer:

```

int a = 10, *ptrX, *ptrY;
ptrX = &a;
ptrY = &ptrX;
*&ptrX = *(&a) = a

```

the value of *&ptrX is the value of a, which is 10.

III. Programming

a. Create a function that will search an element in an array of real numbers and returns the index where the element is located and returns -1 if the element is not found.

Sample function $x = \text{search}(A, E)$; where x gets the index where the E is located in array A .

ANSWER: –

b. Create a function that will sort the list of integers in ascending or descending order.

Sample function $S = \text{sort}(A, E)$; where S is the sorted list, A is the original list, and E is the choice to sort in ascending or descending order

ANSWER: –

c. Create a function that will insert into the list of characters and returns 0 if the insert is unsuccessful and 1 if the insert is successful.

Sample function $x = \text{insert}(A, E)$; where x is either 0 or 1, A is the array of characters and E is the index where the element is to be inserted.

ANSWER: –

d. Create a function that will delete an element from a stack and returns the deleted element.

Sample function $x = \text{pop}(S)$; where x is the deleted element and S is the stack where the element is deleted. However if the stack is empty it should return -1.

ANSWER: –

e. Create a function that will add an element to a queue and return the element that was added or -1 if adding is unsuccessful.

Sample function $x = \text{enqueue}(Q, E)$; where x is the element is being added or -1 otherwise, Q is the queue where E is added.

ANSWER: –

f. Create a function that will transfer all elements of a queue into a stack.

Sample function $S = \text{transfer}(Q)$; where S is the stack of elements and Q is the queue of elements.

ANSWER: –